Data Quality & KPI Dictionary

Technical assurance note | Superstore Power BI model

Source and processing record

Source: Sample - Superstore.csv, 9,994 order-line records and 21 columns. Reporting period: 3 January 2014 to 30 December 2017 by Order Date. Query name: FactSales. Dates parsed using United States month/day/year interpretation. Product Name text trimmed. Extreme values, negative profits and an unresolved near-duplicate were retained pending business-source confirmation.

Quality control	Result	Decision
Missing source values	0	No imputation or incomplete-row deletion
Exact full-row duplicates	0	No blanket deduplication
Near-duplicate ignoring Row ID	1 extra matching line	Retained; requires transactional confirmation
Negative-profit lines	1,871	Retained for profitability assessment
Ship-before-order anomalies	0	No date-order correction required
Product IDs with multiple names	32	Use Product Name for product ranking
Distinct orders / customers	5,009 / 793	Use DISTINCTCOUNT for Order ID

KPI dictionary

Measure	DAX expression	Full-period result
Total Sales	SUM('FactSales'[Sales])	2,297,200.86
Total Profit	SUM('FactSales'[Profit])	286,397.02
Total Orders	DISTINCTCOUNT('FactSales'[Order ID])	5,009
Average Sales	AVERAGE('FactSales'[Sales])	229.86 / line
Profit Margin	DIVIDE([Total Profit], [Total Sales])	12.47%

Model schematic

DimDate[Date] (unique days, 2014–2018) 1 → * FactSales[Order Date] (order-line transactions); active relationship; single cross-filter direction. Fiscal and business calendars were not supplied; analysis uses calendar years and order dates.

Reconciliation controls

Control	Expected full-period result
Total quantity	37,873
Total Sales	2,297,200.8603 before presentation rounding
Total Profit	286,397.0217 before presentation rounding
2014 Sales	484,247.50
2015 Sales	470,532.51
2016 Sales	609,205.60
2017 Sales	733,215.26
Unfiltered report margin	12.47%

Interpretive notes

- Sales is a numeric field, not a payment record; do not infer actual cash collection.
- Source Profit is not reconciled to a formal income statement; do not call it audited net earnings.
- Discount is stored as a decimal fraction (for example, 0.20 = 20%).
- An outlier flag is a screening tool, not evidence of an error or fraud.
- Static PDF reports cannot demonstrate Power BI slicer interactivity; retain the PBIX project as the source interactive artifact.

Repeatable verification procedure

Clear Year and Region slicers. Confirm Total Sales 2,297,200.86, Total Profit 286,397.02, Total Orders 5,009, Average Sales 229.86, and Profit Margin 12.47%. Select year 2017 and confirm Sales 733,215.26. Reset filters before PDF export. Review page labels, chart sorting, map extent, matrix totals and legibility.

Use of this document

This appendix supports the Dataset Inspection and Professionalism & Documentation criteria by recording transformation decisions, model grain, business metric definitions and reconciliation checks. It supplements rather than replaces the required interactive Power BI project.